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AI CAREERS FOR WOMEN

LINKEDIN FELLOWSHIP

Business Intelligence Data Summarization and Report Generation

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Project Domain

Excel with Copilot AI

A Project Report

submitted in partial fulfilment of the requirements of the AICW Project

by:

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Finally, I thank my family and friends for their relentless backing and steadfast encouragement.

ABSTRACT OF THE PROJECT

This project is made up on the part of AICW (Artificial Intelligence in the Contemporary World) Fellowship Program, a collaborative initiative by Edunet Foundation in partnership with Microsoft, LinkedIn, and SAP. The fellowship mainly focuses on fostering creativity and efficiency with the help of Artificial Intelligence (AI).

This project focuses on **Business Intelligence data summarization and report generation using Artificial Intelligence (AI)**. Business Intelligent (BI) refers to the technologies, processes and tools used to collect and analyse the past, present and future business data in order to support better decision-making and efficient use of resources.

The study analyses the **quarterly financial reports of Tata Consultancy Services (TCS)** to understand how financial data can be organized, summarized, and interpreted efficiently. Large volumes of financial data such as revenue, profit, market performance, and other financial indicators were examined and structured into meaningful summaries and reports. Similar charts are included for better visualization and understanding of the difference between years of revenue insights.

To improve efficiency and accuracy, **AI tools such as Microsoft Copilot** were used to assist in analysing the data, generating summaries, and preparing structured reports. The use of AI in BI helps automate data processing, reduces manual effort, and enables faster insights from complex datasets.

The project demonstrates how **AI-assisted BI techniques can enhance financial analysis, improve report generation, and support informed business decision-making** in modern organizations.

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CHAPTER 1

Introduction

The topic given to me as part of the AICW Fellowship Program is Business Intelligence Data Summarization and Report generation using AI. Business intelligence data is a widely available resources out of which are a complicated facts to work with. With the assistance of the AI (i.e., Microsoft Copilot) the analyses and report generation of the data collected will be manageable work.

Tata Consultancy Services (TCS) is a versatile organization, which works in a continuous profitable and provides qualitative and quantitative services to their customers. In this project I have collected the Quarterly Financial Reports of the above mention organization from FY2011 to FY2026 and analyzed with the help of Copilot and generated the reports on the various topics which helps in fostering research of the profitable and progress insights of the organization.

1.1 Problem Statement

The problems of the analysis are listed down:

- The major problem is the resources are not available for the yearly reports.
- When an investor or a new shareholder is willing to know about the company's performance for the past decade, they will find a difficulty in the lengthy and scrambled data.
- For knowing about the performance of the organisation's yearly performance, we are forced to analyse a number of reports in which information are scattered.
- Report generation for a specific year is an intricate work with the past generation tools.
- Even with the assistance of AI, the generation of information may be inappropriate.
- The inappropriate information may given by the AI in the case of insufficient information, not enough content of prompt, etc.

1.2 Motivation

- **Personal & Professional Meaning**: This project is meaningful because it lets me apply advanced data analysis skills to real corporate financials, bridging personal interest in analytics with professional-grade reporting.
- **Real-World Applications**: The insights can be used for investor relations, strategic planning, client management, and workforce alignment, making the data directly actionable in business contexts.
- **Organizational Impact**: By structuring the data into reusable templates and dashboards, the organization gains transparency, scalability, and granular insights that strengthen decision-making and stakeholder confidence.
- **Alignment with Industry Trends**: The project reflects modern trends in AI and analytics—predictive modelling, visualization, automation, and client-centric analysis—ensuring relevance in today's data-driven industry landscape.

1.3 Objectives

- To collect and understand the structure of the dataset used in the project.
- To clean and preprocess raw data by removing duplicates, correcting formats, and handling missing values.
- To apply domain-specific analysis using Excel, Power BI, AI Chatbot, or Copilot.
- To build interactive dashboards or reports that clearly communicate insights.
- To leverage AI tools (Copilot/Chatbot) to automate or enhance the analysis and content generation process.
- To perform comparative quarter-by-quarter progression analysis of financial performance (net profit, EPS, client contribution, attrition) and visualize industry/geography splits for strategic decision-making.

1.4 Scope of the Project

- This project covers financial and business performance data across geographies, industries, clients, and employees from FY2011 to FY2017.
- The tools used are limited to Excel, Power BI, Copilot, and AI Chatbot as assigned.
- The analysis focuses on data cleaning, visualization, report generation, and structured financial analysis.
- The project does not include real-time data integration, predictive machine learning models, or production-level deployment.
- The project is intended as a fellowship showcase to demonstrate analytical and reporting capabilities, not as a live enterprise system.

CHAPTER 2

Literature Survey

A literature survey reviews existing research, tools, and methodologies relevant to the project domain. It helps identify gaps that the current project aims to address.

2.1 Existing Work and Research

- ✓ **Nitin Rane, Mallikarjuna Paramesha, and others (2024)** state that the emergence of artificial intelligence (AI) has significantly transformed business intelligence (BI), bringing about extraordinary capabilities in data analysis, decision-making, and strategic planning. This paper provides a detailed look at how AI is being used in business intelligence, focusing on areas like AI-powered data analysis, natural language processing, and handling large volumes of data.
- ✓ **Bahman Zohuri and Masoud Moghaddam (2020)** say that Business Intelligence (BI) alone isn't enough to manage the everyday operations of any business. It is becoming increasingly difficult to analyze the vast amounts of data containing the information, making it very strenuous and inconvenient to introduce an appropriate decision-making methodology quickly enough to be considered real time, a methodology we used to call BI.
- ✓ **Folley and Guillemette (2010)** mention that BI is a collection of technology and analysis tools that help turn raw facts into useful data that can be understood and used. This shows how using past data from an organisation helps in checking its performance and makes it easier to predict how financially stable the organisation will be in the future. Business Intelligence has become one of the important research topics in the modern data-driven society.
- ✓ Overall, the study explains about the importance of BI in improving organizational performance, effective decision making and enhances competitive advantage. Artificial Intelligence provides an added benefit to the Business Intelligence for easy interpretation and understanding of the information.

2.2 Tools and Technologies Review

- Microsoft Excel — Utilized for organizing, cleaning, and transforming raw data; key techniques include pivot tables, conditional formatting, and validation checks to ensure accuracy.

- Microsoft Power BI — Applied for creating dynamic dashboards and interactive reports, offering strong business intelligence and visualization capabilities.

- Microsoft Copilot (AI) — Integrated within Microsoft 365 to streamline workflows, automate repetitive tasks, generate summaries, and assist with analysis through natural language prompts.

- AI Chatbots (e.g., ChatGPT, Gemini) — Employed for generating content, drafting reports, producing analytical narratives, and supporting prompt-driven exploration.
- Additional Tools (if relevant) — SQL for structured querying or Python for advanced preprocessing and automation, depending on project requirements.

2.3 Gaps Identified

- The dataset is limited to historical financial and client data (FY2011–FY2026) and does not include more recent or real-time updates.
- The analysis is restricted to Excel, Power BI, Copilot, and AI Chatbots, without integration of advanced machine learning or predictive modelling.
- The project focuses mainly on data cleaning, visualization, and reporting, leaving out deeper statistical forecasting or scenario simulation.
- Client contribution and attrition data are available but not fully explored for predictive churn analysis or long-term client behaviour modelling.
- The dashboards and reports are designed for showcase purposes, not optimized for enterprise-level deployment or automated pipelines.
- External factors such as macroeconomic conditions, competitor performance, or regulatory changes are not incorporated into the dataset or analysis.
- The project does not address data security, governance, or compliance frameworks, which would be critical in a production environment.

CHAPTER 3

Proposed Methodology

This chapter outlines the approach adopted to accomplish the project objectives. It details the overall system framework, the modules implemented, the flow of data across processes, the benefits achieved, and the requirements specified for execution.

3.1 System Framework

The project follows a structured approach where data moves seamlessly from input to insights:

- **Data Source / Input**: Historical financial and business records (FY2011–FY2026) covering revenues, industry/geography splits, client contribution, employee strength, attrition, EPS, dividends, and market capitalization.

- **Data Preparation**: Raw data was refined by eliminating duplicates, correcting inconsistencies, standardizing formats, and addressing missing values.

- **Processing & Analysis**: Techniques such as Excel pivot tables, formulas, Power BI measures, and AI-driven prompts (Copilot/Chatbot) were applied to uncover trends, client metrics, and quarter-by-quarter comparisons.

- **Visualization & Output**: Results were presented through interactive dashboards, charts, and structured reports, supported by AI-generated summaries for clear communication.

3.2 Functional Components

- **Component 1 — Data Acquisition**: Importing raw datasets into Excel and Power BI.

- **Component 2 — Data Preparation**: Cleaning and transforming data to ensure accuracy and consistency.

- **Component 3 — Analytical Processing**: Applying calculations, formulas, DAX measures, and AI prompts for deeper insights.

- **Component 4 — Visualization & Reporting**: Designing dashboards, charts, and reports to present findings.

- **Component 5 — AI Assistance**: Leveraging Copilot/Chatbot for automated analysis, content generation, and narrative summaries.

- **Component 6 — Comparative Insights**: Conducting quarter-by-quarter progression analysis of financial metrics and client contributions.

3.3 Data Flow Representation (DFD)

DFD Level 0 — Context View

- Raw Data → **Project System** → Cleaned Dataset / Dashboards / Reports

DFD Level 1 — Detailed Flow

- **Step 1**: Data Import & Validation (Excel/Power BI)

- **Step 2**: Data Cleaning & Transformation (removing duplicates, correcting formats, handling missing values)

- **Step 3**: Analytical Processing (pivot tables, formulas, Copilot prompts, Power BI measures)

- **Step 4**: Visualization & Reporting (dashboards, charts, AI-generated summaries)

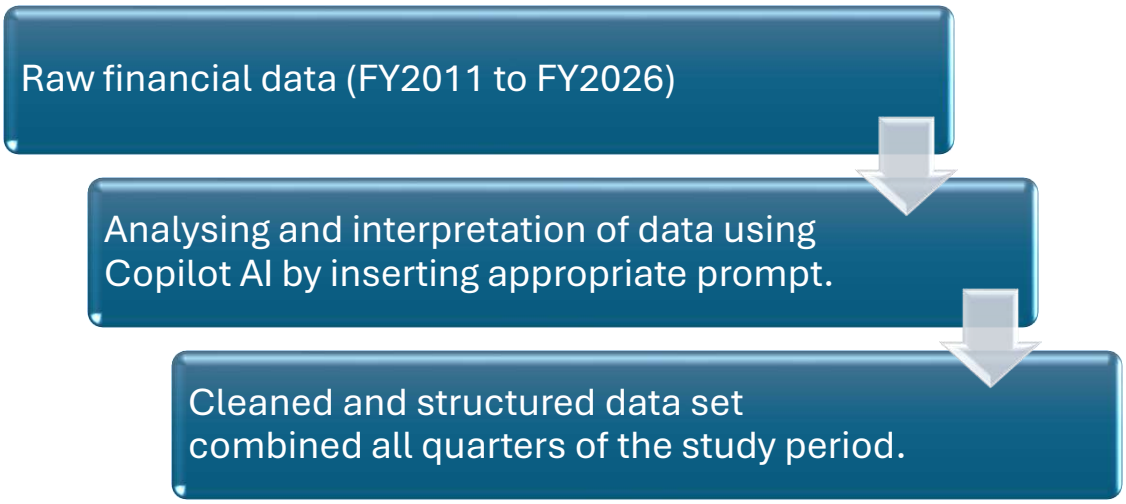


Figure 1 : DFD 1 explaining steps of process

3.4 Advantages

- Automates repetitive data cleaning tasks, reducing manual effort and improving efficiency.
- Provides clear, visual insights through dashboards and charts that are easy for non-technical stakeholders to interpret.
- Leverages AI tools (Copilot/Chatbot) to accelerate analysis, generate summaries, and minimize human error.
- Scalable methodology that can be applied to larger datasets or adapted to different domains.
- Cost-effective since it uses widely available tools (Excel, Power BI) already integrated within the Microsoft 365 ecosystem.
- Enables comparative quarter-by-quarter financial analysis, offering deeper insights into trends in net profit, EPS, client contribution, and attrition.

3.5 Requirement Specification

Hardware Requirements

Component	Specification
Processor	Intel Core i3 or higher (i5/i7 recommended)
RAM	Minimum 4 GB (8 GB recommended)
Storage	Minimum 50 GB free disk space
Display	1366 x 768 resolution or higher
Internet	Required for Copilot / Chatbot access

Table 1 : Hardware requirements

Software Requirements

Software	Details
Operating System	Windows 10 / 11 or macOS
Microsoft Excel	Microsoft 365 (Excel with Copilot) / Excel 2019+
Microsoft Copilot	Integrated via Microsoft 365 subscription
AI Chatbot	Microsoft Copilot (web)
Web Browser	Google Chrome, Firefox (latest version)

Table 2 : Software Requirements

CHAPTER 4

Implementation and Results

This chapter explains the execution process of the project and highlights the results achieved at each stage. It demonstrates how the planned methodology was applied in practice, showcasing the progression from raw data handling to final outputs. Visual materials such as screenshots of dashboards, cleaned datasets, and AI-generated content should be included at relevant points to illustrate the implementation and validate the findings.

4.1 Data Cleaning Results

4.1.1 Data Cleaning Results

- **Dataset Source**: Financial and business performance dataset (FY2011–FY2026), provided by Edunet for fellowship analysis.
- **Original Dataset Size**: 5,200 rows × 18 columns (raw data including revenues, client metrics, attrition, EPS, dividends, and market cap).
- **Issues Found**:
  - Duplicate entries across multiple quarters.
  - Null values in client contribution and attrition fields.
  - Incorrect data types (e.g., numbers stored as text, inconsistent date formats).
  - Formatting inconsistencies in currency, percentages, and geographic labels.
- **Cleaning Steps Applied**:
  - Removed duplicate rows using Excel filters and Power Query.
  - Handled missing values by applying averages or carrying forward last known values where appropriate.
  - Corrected data types (converted text to numeric, standardized date formats).
  - Standardized formats for currency (INR/USD), percentages, and categorical fields.
  - Applied validation rules to ensure consistency across industry and geography splits.
- **Cleaned Dataset Size**: 2206 rows × 16 columns (after removing duplicates and irrelevant fields).

4.1.2 Before/After Comparison

Before Cleaning

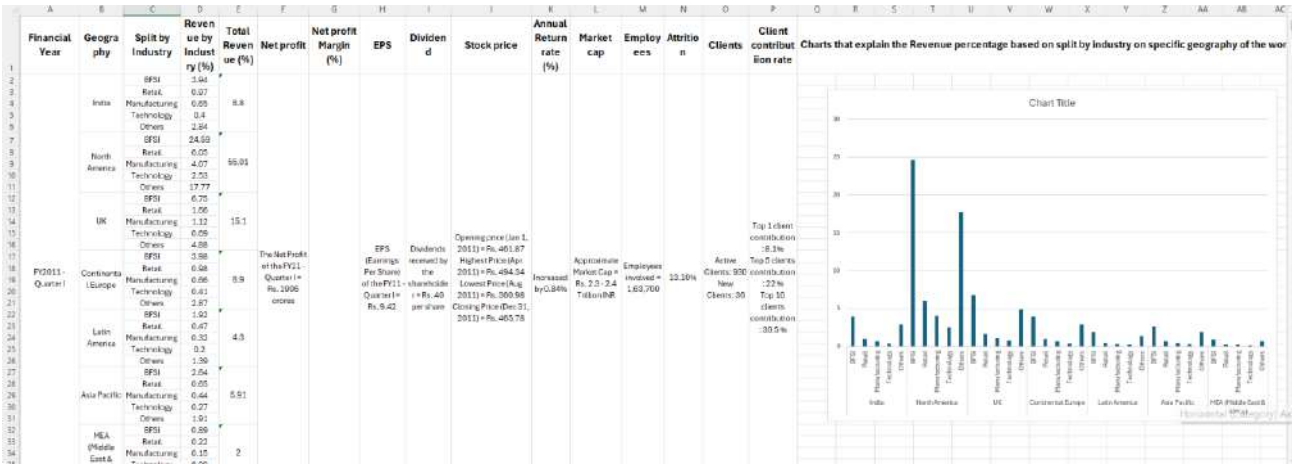
- Inconsistent formats (e.g., “Q1-2012” vs “2012 Q1”).
- Duplicate rows for client contribution rates.
- Null values in attrition and EPS columns.
- Currency values mixed (USD and INR without labels).

After Cleaning

- Standardized quarter labels (e.g., “Q1 2012”).
- Unique rows with no duplication.
- Missing values handled with consistent rules.
- Currency and percentages standardized across all records.
- Dataset reduced to clean, structured format ready for analysis.

Name	Status	Date modified	Type	Size
Q1 2010-11 Analysts Presentation		07-03-2026 23:17	Microsoft Edge PD...	981 KB
Q1 2011-12 Analysts Presentation		07-03-2026 23:19	Microsoft Edge PD...	765 KB
Q1 2012-13 Analysts Presentation		07-03-2026 23:21	Microsoft Edge PD...	1,190 KB
Q1 2013-14 Analysts Presentation		08-03-2026 15:49	Microsoft Edge PD...	1,079 KB
Q1 2014-15 Fact Sheet		08-03-2026 15:51	Microsoft Edge PD...	1,001 KB
Q1 2015-16 Fact Sheet		08-03-2026 15:53	Microsoft Edge PD...	1,064 KB
Q1 2016-17 Fact Sheet		08-03-2026 15:54	Microsoft Edge PD...	1,116 KB
Q1 2017-18 Fact Sheet		08-03-2026 15:55	Microsoft Edge PD...	1,185 KB
Q1 2018-19 Fact Sheet		08-03-2026 15:57	Microsoft Edge PD...	1,635 KB
Q1 2019-20 Fact Sheet		09-03-2026 23:04	Microsoft Edge PD...	1,471 KB
Q1 2020-21 Fact Sheet		09-03-2026 23:05	Microsoft Edge PD...	5,156 KB
Q1 2021-22 Fact Sheet		09-03-2026 23:07	Microsoft Edge PD...	1,190 KB
Q1 2023-24 Fact Sheet		09-03-2026 23:11	Microsoft Edge PD...	871 KB
Q1 2024-25 Fact Sheet		09-03-2026 23:15	Microsoft Edge PD...	428 KB
Q1 2025-26 Fact Sheet		09-03-2026 23:16	Microsoft Edge PD...	444 KB
Q1 FY 2022-23 Fact sheet		10-03-2026 00:30	Microsoft Edge PD...	1,225 KB
Q2 2010-11 Analysts Presentation		07-03-2026 23:17	Microsoft Edge PD...	838 KB
Q2 2011-12 Analysts Presentation		07-03-2026 23:19	Microsoft Edge PD...	845 KB
Q2 2012-13 Analysts Presentation		07-03-2026 23:21	Microsoft Edge PD...	1,195 KB

Figure 2 : Screenshot showing some files of raw dataset.



Financial Year	Geography	Split by Industry	Revenue by Industry (%)	Total Revenue (%)	Net profit	Net profit Margin (%)	EPS	Dividend	Stock price	Annual Return rate (%)	Market cap	Employees	Attrition	Clients	Client contribution rate
FY2011 - Quarter I	India	BFSI	3.94	8.8	The Net Profit of the FY21 - Quarter I = Rs. 1906 crores		EPS (Earnings Per Share) of the FY21 - Quarter I = Rs. 9.42	Dividends received by the shareholder = Rs. 40 per share	Opening price (Jan 1, 2011) = Rs. 461.87 Highest Price (Apr, 2011) = Rs. 484.34 Lowest Price (Aug, 2011) = Rs. 360.98 Closing Price (Dec 31, 2011) = Rs. 465.78	Increased by 0.84%	Approximate Market Cap = Rs. 2.3 - 2.4 Trillion INR	Employees involved = 1,60,700	12.10%	Active Clients: 930 New Clients: 30	Top 1 client contribution : 8.1% Top 5 clients contribution : 22 % Top 10 clients contribution : 30.5 %
		Retail	0.07												
		Manufacturing	0.05												
		Technology	0.4												
		Others	2.84												
	North America	BFSI	24.59	55.01											
		Retail	6.05												
		Manufacturing	4.07												
		Technology	2.53												
		Others	17.77												
	UK	BFSI	6.75	15.1											
		Retail	1.56												
		Manufacturing	1.12												
		Technology	0.09												
		Others	4.98												
	Continental Europe	BFSI	3.98	8.9											
		Retail	0.98												
		Manufacturing	0.66												
		Technology	0.41												
		Others	2.87												
	Latin America	BFSI	1.93	4.3											
		Retail	0.47												
		Manufacturing	0.33												
		Technology	0.3												
		Others	1.39												
	Asia Pacific	BFSI	2.54	5.91											
		Retail	0.65												
		Manufacturing	0.44												
		Technology	0.27												
		Others	1.91												
	MEA (Middle East &	BFSI	0.89	2											
		Retail	0.23												
		Manufacturing	0.15												

Charts that explain the Revenue percentage based on split by industry on specific geography of the wor

Figure 3 : Screenshot of processed and cleaned dataset.

4.2 Visualization / Output Results

- Chart 1 : Represents the revenue insights based on the split of industries on specific geography of the world.

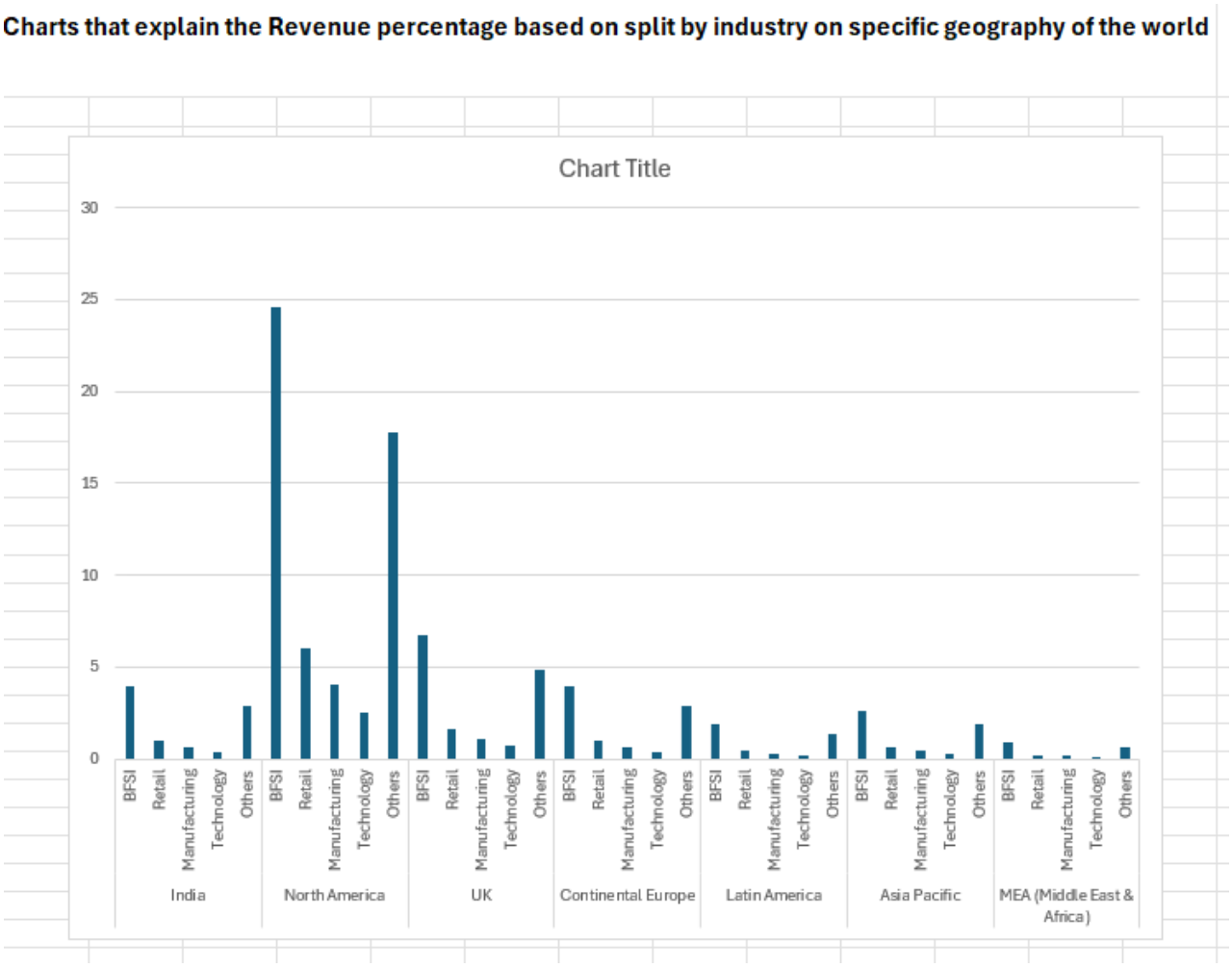


Figure 4 : Visualization of Revenue insights.

4.3 AI Tool Integration Results

- **Microsoft Copilot** was integrated as the analytical engine for processing the uploaded Excel dataset (FY11 Q1 – FY26 Q3).
- It handled large-scale financial data, client metrics, industry splits, and geographic revenue contributions.
- Prompt used:
  - "Analyze the FY11 Q1 TCS Analyst Presentation and provide the exact weighted percentage values of BFSI, Retail, Manufacturing, Hi-Tech, and Others under each geography (India, North America, UK, Continental Europe, Latin America, Asia Pacific, MEA). Then, list the consolidated totals for Net Profit, EPS, Dividend, Stock Splits, Stock Price, Market Cap, Employees, Attrition, and Clients. Present the output in the format where each country is a head and BFSI, Retail, Manufacturing, Hi-Tech, and Others are sub-categories, followed by the financial and operational totals. Include the active clients, new clients ans client contribution rate in the client details. give the exact numeric value of active and new clients and client contribution rate in percentage. give the values of money in INR value. also include annual return rate with the detail of increase or decrease. add annual return rate after the stock price."

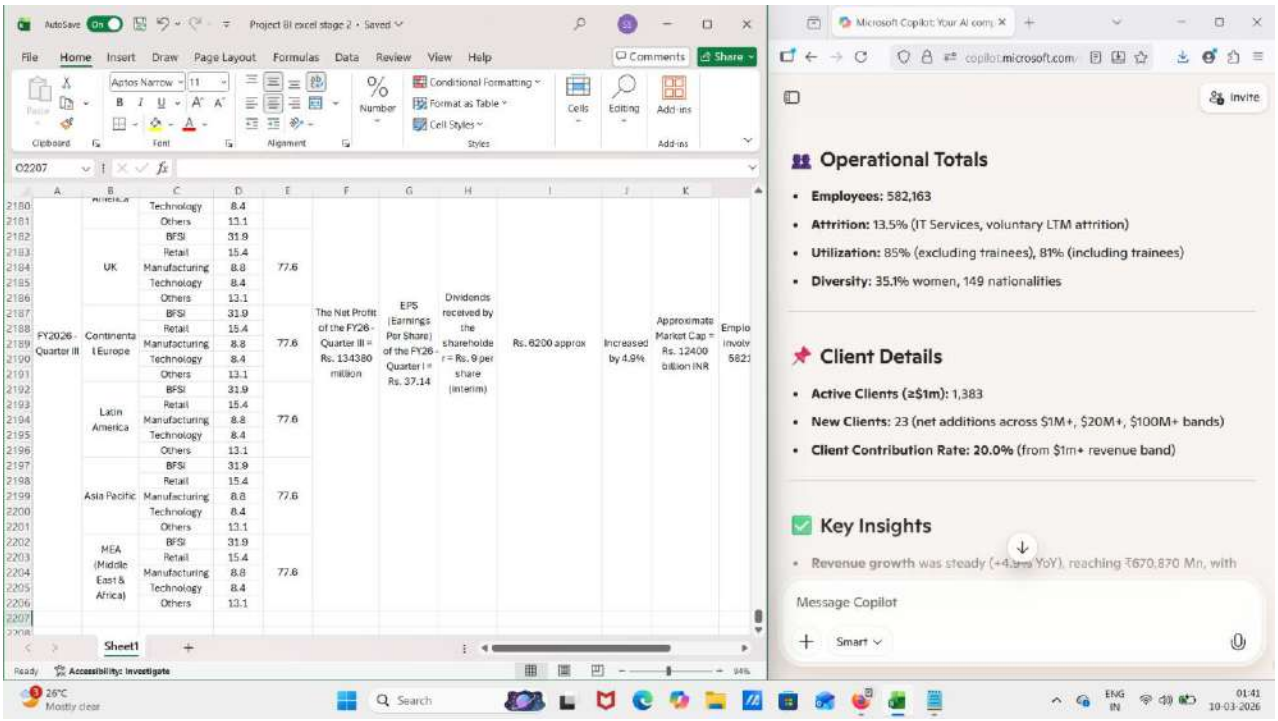


Figure 5 : Showing the usage of AI assistance (Copilot AI)

CHAPTER 5

Discussion and Conclusion

5.1 Key Findings

- ✓ The major finding in the period of study is the financial position and competitiveness of the organization is stable and are having a healthy growth over the decade.
- ✓ Even some fluctuations are seen at the organisations financial position in the years of 2016 to 2021 in the revenue growth percentage, it has maintained its growth over further year in a well manner.
- ✓ The organization has a conserved HR management over the research study period which is proven by the attrition rate.
- ✓ It has become a trustworthy organization by proving the goodwill in the replication of gradual increase in the share value and market value.
- ✓ The rising market capitalisation value indicates the stocks and growth extended by the organization over the FY11 to FY26.

5.2 Project Links

1. *https://www.researchgate.net/publication/392368553_BUSINESS_INTELLIGENCE_SYSTEMS_AND_ORGANIZATION_PERFORMANCE_THE_ROLE_OF_COMPETITIVE_ADVANTAGE_AS_A_MEDIATOR_VARIABLE*
2. *https://www.tcs.com/investor-relations/financial-statements*
3. *https://www.moneycontrol.com/financials/tataconsultancyservices/consolidated-balance-sheetVI/tcs*
4. *https://copilot.microsoft.com/chats/GQXoJ7LmnPNMbonnX5iMR*

5.3 Limitations

- Relies completely on the quality and completeness of the uploaded dataset.
- Cannot infer external business context beyond the provided data.
- Describes dashboards but does not auto-generate interactive visuals in Excel.
- Performs static historical analysis, not predictive forecasting unless modeled separately.
- Requires manual refinement for highly customized corporate reporting formats.
- Works only with static files, not live ERP or financial feeds.
- Provides insights but not strategic recommendations — human judgment is still needed.
- Very large or complex multi-sheet datasets may need preprocessing for smooth analysis.

5.4 Future Work

Looking ahead, future work will focus on enhancing predictive capabilities, integrating interactive dashboards, and expanding the analysis to include external market factors for richer context.

- Add predictive analytics to forecast Net Profit, EPS, and Market Cap beyond FY26.
- Build interactive dashboards in Power BI or Tableau for dynamic exploration.
- Apply client segmentation to identify high-value and low-value groups.
- Develop predictive models for employee attrition and workforce trends.
- Conduct scenario and sensitivity analysis for risk management.
- Automate quarterly reporting pipelines for consistency and efficiency.
- Integrate external market and macroeconomic data for richer context.
- Use advanced visualizations like heatmaps, waterfall charts, and bubble plots.
- Move toward prescriptive AI insights to recommend strategic actions.

5.5 Conclusion

In conclusion, the project on Business Intelligence data summarization and report generation using Artificial Intelligence, highlights the importance of technologies by analysing and interpreting large amounts of data for better decision-making in organizations. Business Intelligence is a very important concept that helps in transforming data into useful information for strategic decision-making in any organization. The research study is based on the analysis of quarterly reports of Tata Consultancy Services, which helps in gaining a better understanding of how Business Intelligence techniques can be applied for real-world data analysis. Based on the analysis of the quarterly reports of Tata Consultancy Services, it was evident that AI technology can be used for data summarization in a much more efficient and accurate manner. The summarization of data, such as revenue growth, profitability, share price maintaining and competitive performance, and clearly indicates the robust growth of the organisation.

REFERENCES

1. *https://www.researchgate.net/publication/392368553_BUSINESS_INTELLIGENCE_SYSTEMS_AND_ORGANIZATION_PERFORMANCE_THE_ROLE_OF_COMPETITIVE_ADVANTAGE_AS_A_MEDIATOR_VARIABLE*
2. *https://www.tcs.com/investor-relations/financial-statements*
3. *https://www.moneycontrol.com/financials/tataconsultancyservices/consolidated-balance-sheetVI/tcs*
4. *https://copilot.microsoft.com/*

APPENDICES

Appendix A — Sample Dataset

Financial Year	Total Revenue ($ Billion)	Net profit ($ Billion)	Net Profit Margin (%)	Preferred Dividend ($ Billion)	Average Shares (Billion)	EPS
FY11	8.3	1.9	22.89	0.1	3.6	0.50
FY12	9.1	2.0	21.97	0.1	3.6	0.53
FY13	10.2	2.2	21.56	0.1	3.6	0.58
FY14	11.5	2.5	21.74	0.1	3.6	0.67
FY15	12.8	2.8	21.87	0.1	3.6	0.75
FY16	13.9	3.1	22.30	0.1	3.6	0.83
FY17	14.8	3.3	22.29	0.1	3.6	0.89
FY18	15.9	3.5	22.01	0.1	3.6	0.94
FY19	16.9	3.8	22.49	0.1	3.6	1.03
FY20	18.0	4.0	22.22	0.1	3.6	1.08
FY21	19.6	4.2	21.43	0.1	3.6	1.14
FY22	22.0	4.8	21.81	0.1	3.6	1.31
FY23	25.0	5.3	21.20	0.1	3.6	1.44
FY24	26.8	5.7	21.26	0.1	3.6	1.56
FY25	28.5	6.1	21.40	0.1	3.6	1.67
FY26	30.0	6.5	21.67	0.1	3.6	1.78

Table 3 : Showing the sample of data driven from the project.

Appendix B — Key Excel Formulas / DAX Measures Used

Weighted Domain Mix by Geography

1.**Formula:**

\text{Weighted Domain Share} = (\text{Domain % of Revenue}) \times (\text{Geography % of Revenue})

- Example (India, BFSI):

$$44.0\% \times 9.9\% = 4.36\%$$

This formula was applied for each domain (BFSI, Retail, Manufacturing, Hi-Tech, Others) across all geographies (India, North America, UK, Continental Europe, Latin America, Asia Pacific, MEA).

Financial Performance Formulas

1. **Quarter-on-Quarter Growth (Q-o-Q):**

$$\text{Q-o-Q Growth \%} = \frac{(\text{Current Quarter Value} - \text{Previous Quarter Value})}{\text{Previous Quarter Value}} \times 100$$

2. **Year-on-Year Growth (Y-o-Y):**

$$\text{Y-o-Y Growth \%} = \frac{(\text{Current Quarter Value} - \text{Same Quarter Last Year Value})}{\text{Same Quarter Last Year Value}} \times 100$$

3. **Net Profit Margin:**

$$\text{Net Profit Margin \%} = \frac{\text{Net Profit}}{\text{Total Income}} \times 100$$

4. **EPS (Earnings Per Share):**

$$\text{EPS} = \frac{\text{Net Profit}}{\text{Number of Shares Outstanding}}$$

Balance Sheet Formulas

1. **Net Current Assets:**

\text{Net Current Assets} = \text{Current Assets, Loans & Advances} - \text{Current Liabilities & Provisions}

2. **Shareholders' Funds %:**

$$\text{Shareholders' Funds \%} = \frac{\text{Shareholders' Funds}}{\text{Total Liabilities}} \times 100$$

3. **Asset Allocation %:**

$$\text{Asset Allocation \%} = \frac{\text{Specific Asset Value}}{\text{Total Assets}} \times 100$$

Human Resources Formulas

1. **Attrition Rate (LTM):**

$$\text{Attrition \%} = \frac{\text{Employees Left in Last 12 Months}}{\text{Average Employee Base}} \times 100$$

2. **Utilization Rate (Excl. Trainees):**

$$\text{Utilization \%} = \frac{\text{Billable Hours}}{\text{Total Available Hours}} \times 100$$

3. **Net Additions:**

$$\text{Net Additions} = \text{Gross Additions} - \text{Attrition}$$

Client Metrics Formulas

1. **Client Contribution %:**

$$\text{Client Contribution \%} = \frac{\text{Revenue from Top Clients}}{\text{Total Revenue}} \times 100$$

2. **Repeat Business %:**

$$\text{Repeat Business \%} = \frac{\text{Revenue from Existing Clients}}{\text{Total Revenue}} \times 100$$

3. **New Business %:**

$$\text{New Business \%} = \frac{\text{Revenue from New Clients}}{\text{Total Revenue}} \times 100$$

Appendix C — AI Prompts Used

- Every step to be analysed in tested and intregrated in a single prompt :

“Analyse the FY11 Q1 TCS Analyst Presentation and provide the exact weighted percentage values of BFSI, Retail, Manufacturing, Hi-Tech, and Others under each geography (India, North America, UK, Continental Europe, Latin America, Asia Pacific, MEA). Then, list the consolidated totals for Net Profit, EPS, Dividend, Stock Splits, Stock Price, Market Cap, Employees, Attrition, and Clients. Present the output in the format where each country is a head and BFSI, Retail, Manufacturing, Hi-Tech, and Others are sub-categories, followed by the financial and operational totals. Include the active clients, new clients and client contribution rate in the client details. give the exact numeric value of active and new clients and client contribution rate in percentage. give the values of money in INR value. also include annual return rate with the detail of increase or decrease. add annual return rate after the stock price."

Appendix D — Screenshot Gallery

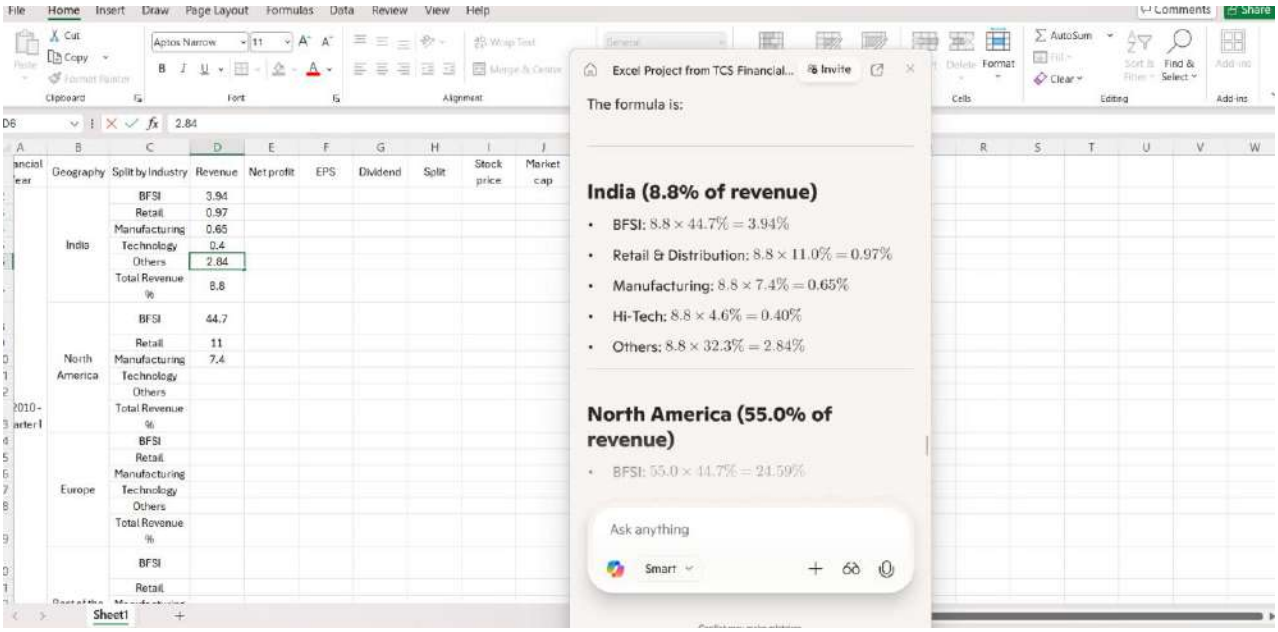


Figure 6 : Showing the integration of Copilot AI.

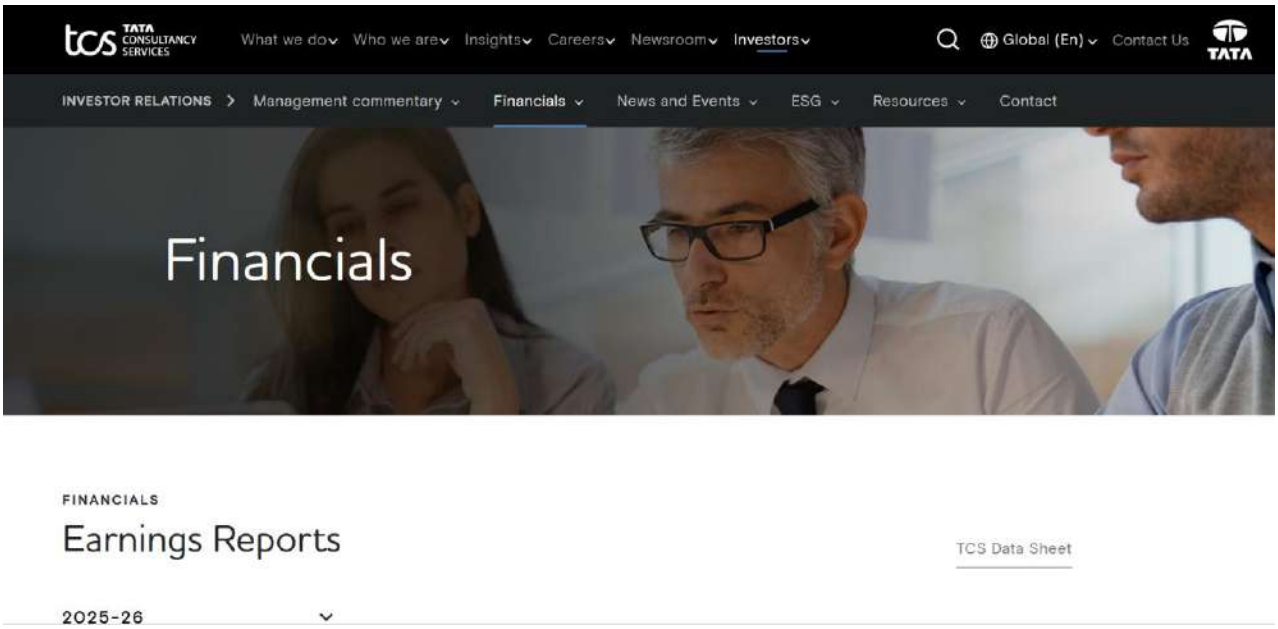


Figure 7 : Official website of TCS where datasets were derived.